

Colin Zeng

919-360-5960 | colin.zeng@duke.edu | [LinkedIn](#) | [Github](#)

EDUCATION

Duke University

GPA: 3.925/4.0

Bachelor of Science in Computer Science and Statistics

Expected May 2028

- Relevant Coursework: Machine Learning, Computer Systems, Database Systems, Data Structures and Algorithms, Multivariable Calculus, Linear Algebra, Advanced Probability, Regression Analysis
- Extracurriculars: Catalyst, Applied Machine Learning, Statistical Science Majors Union
- Awards: U20 World Champion, Team USA, World Junior Ultimate Championships

EXPERIENCE

Undergraduate Research Intern

Jan. 2026 – Present

Dietrich Lab @ Duke University

Durham, NC

- Designed a PostgreSQL schema mapping 1,482 yeast strains across core genes and pangenome traits
- Built a full stack frontend (React, JavaScript, HTML/CSS) with SQL backed queries loading 10,000+ genes across *Saccharomyces cerevisiae* and *Ashbya gossypii* pangenomes (pondslime.ccn.duke.edu)
- Used AlphaFold and BLAST for protein prediction and sequence alignment in the mapping pipeline

Software Engineer Intern

Oct. 2024 - Aug. 2025

Quantworks/SIZEO

Durham, NC

- Helped build a web platform using Anthropic's Claude API to optimize budget allocation, packaging, and seasonal inventory predictions for retail clients including JCPenney and Aeropostale
- Built backend data processing pipelines in Python, visualized with Streamlit and Plotly dashboards
- Designed UI dashboards in Figma for clients and built an artifact gallery storing LLM outputs for analysis

Research Intern

Summer 2023 - Summer 2024

NCSSM Summer Research Programs

Durham & Wilmington, NC

- Presented research on agent-based neuron modeling, real gas behavior, and a modified cholera transmission model incorporating vaccination effects using Mathematica, STELLA Architect, Gaussian, NetLogo, and Maple

PROJECTS

Speech Dysfluency Detection and Removal | *Python, Whisper, scikit-learn*

- Built an end-to-end pipeline to detect, categorize, and remove speech dysfluencies (e.g., repetitions)
- Used OpenAI Whisper to transcribe raw audio and rule-based logic to flag and categorize dysfluency types
- Benchmarked a custom-trained random forest classifier against a premade model for stutter removal accuracy

Insight, Journal Reflection App | *React, Groq API, LLM Integration*

- Built a React web app that generates personalized reflective prompts by feeding journal entries to an open-weight LLM (gpt-oss-20b) via the Groq API
- Designed a from-scratch retrieval system using vectorization and cosine similarity to surface relevant entries
- Implemented client-side data isolation using SHA-256 hashed API keys to namespace user entries without storing raw credentials

Ultimate Frisbee Play Classifier | *PyTorch, YOLOv8, Computer Vision*

- Fine-tuned a YOLOv8 model on a labeled field-photo dataset, achieving 85% precision and 87% recall
- Converted detections into simplified dot-map representations to train a downstream play classifier
- Benchmarked ResNet18, a self-built CNN, and EfficientNet-B0 for play classification; ResNet18 achieved 91.3% accuracy at 1/36th the model size of the custom CNN

Event Horizon, Unreal Engine 5 Game | *C++, Unreal Engine 5, Blueprints*

- Designed a personal game project to learn low-level systems programming and game development workflows
- Built gameplay systems in Unreal Engine 5 (kooling.itch.io) using C++ components with Blueprint scripting

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript/TypeScript, React, R, C++, HTML/CSS

Frameworks & Libraries: React, PyTorch, YOLOv8, scikit-learn, pandas, NumPy

APIs & AI Integration: Groq API, Anthropic Claude API, Whisper, LLM Prompt Engineering

Developer Tools: Git, Unix/Linux, Figma, Roboflow, LaTeX, Google Colab, Plotly, Streamlit